Pressure Injuries: Aetiology and the State of the Art in Detection, Prevention and Treatment

585.617 Rehabilitation Engineering | Module 6 Research Paper | [Your Name] | [Date]

I. Introduction

A pressure injury, still widely called a bedsore or pressure ulcer, is localised damage to skin and underlying soft tissue, usually over a bony prominence or beneath a medical device. The populations affected share one characteristic: an inability to relieve their own loading. These include older adults with reduced mobility, intensive-care and surgical patients, nursing-home residents, people with spinal cord injury, and full-time wheelchair users. The burden is large and concentrated. A Markov simulation of United States hospital-acquired pressure injuries estimated a national cost near $26.8 billion annually, approximately $10,708 per affected patient, with Stage III and IV full-thickness wounds accounting for roughly 58% of that total despite being a minority of cases [1].

Why they arise has been substantially rewritten in the last two decades. The classical account was ischaemic: sustained pressure occludes capillary flow, tissue is starved, and necrosis follows. Contemporary consensus reverses the ordering. Gefen and colleagues describe a damage cascade in which **sustained mechanical deformation of cells is the primary insult**, inflammatory and oedema damage is secondary, and ischaemia is tertiary [2]. Cell death can begin within minutes of high deformation even when oxygen and glucose supply are normal, although hours of loading may pass before anything is clinically visible. The same review distinguishes superficial damage, driven mainly by shear, from deep injury, which arises from high pressure combined with shear over bony prominences or under stiff devices [2]. Deep tissue injury therefore initiates subcutaneously and propagates outward, so by the time it is visible at the skin, irreversible damage has already occurred.

That last fact organises this review. The field's reference standard — a visible change in the skin — is a lagging indicator of a mechanical event that happened earlier and deeper. Detection, prevention trials and staging all inherit that lag, and much of what follows is best understood as the clinical and engineering consequences of measuring an effect rather than its cause.

II. Detection Methods

Standard practice remains visual and tactile skin assessment combined with a risk-stratification instrument such as the Braden Scale, with injuries then staged by depth according to international guideline criteria [3]. The approach is cheap, requires no equipment, and has two structural weaknesses.

The first is timing. Non-blanchable erythema, the Stage 1 criterion, is by definition a surface observation, and for deep tissue injury the subcutaneous damage precedes it [2]. Assessment thus reliably identifies injuries at the point where prevention has already failed. The second is equity. Erythema and blanching response are colour phenomena, and both are substantially harder to observe in darkly pigmented skin; Stage 1 and deep tissue injuries are consequently under-detected and present later in patients with darker skin tones. Guideline recommendations to compare the site against adjacent skin and to palpate for temperature and firmness are partial mitigations for what is fundamentally a limitation of the measurand [3].

Sub-epidermal moisture (SEM) measurement is the most clinically mature alternative. The technique measures biocapacitance changes caused by localised oedema beneath intact skin. In a blinded, prospective, multi-site study of 182 patients across twelve acute and post-acute sites, SEM changes preceded a visually diagnosed pressure injury by a mean of 4.7 ± 2.4 days, with sensitivity of 87.5% but

specificity of only 32.9% [4]. That specificity invites misreading. The comparator is a lagging standard, so a patient flagged by SEM who never develops a visible ulcer is scored as a false positive even when the flag was correct and prompted the intervention that prevented the wound; the authors note this confounding directly [4]. From an engineering standpoint the relevant claim is not diagnostic accuracy against visual assessment but lead time, and roughly five days of it is substantial. SEM also measures a tissue property rather than a colour, which in principle makes it independent of skin tone — an equity argument as much as a technical one. Thermography, ultrasound and interface pressure mapping are investigated on similar logic, with less clinical evidence behind them.

III. Prevention Methods

Prevention rests on three pillars: redistributing load, managing the skin microclimate, and maintaining nutrition and tissue tolerance [3]. The striking feature of the evidence base is how little of it is certain.

Manual repositioning, conventionally on a two-hourly schedule, is the oldest intervention and is examined in a Cochrane review of repositioning frequency and body position that was unable to establish the superiority of any particular regimen [5]. Support surfaces fare similarly: a Cochrane review of surfaces for prevention found the comparison between higher-specification constant-low-pressure and alternating-pressure surfaces unclear, although specialised foam outperformed the standard hospital mattress [6]. A later review restricted to alternating-pressure (active) air surfaces pooled four trials with 2,247 participants and reported a risk ratio of 0.63 (95% CI 0.34 to 1.17) against foam — a point estimate suggesting benefit, a confidence interval including no effect, and low-certainty evidence [7]. Prophylactic dressings and topical agents are similarly inconclusive as to which, if any, is most effective [8].

It would be wrong to read this as evidence that prevention does not work; it is better read as evidence that the trials are heterogeneous, underpowered, and measuring the wrong endpoint. When the outcome is a visible ulcer, every trial is waiting for a late, relatively rare event, which is precisely the design that produces wide confidence intervals. A measurable subclinical endpoint of the kind SEM provides would let prevention trials detect differences long before a wound forms, and this methodological argument may matter more than any individual device comparison.

IV. Treatment Methods

Once an injury forms, treatment is staged. Superficial injuries are managed by offloading, moisture balance and protective dressings; full-thickness wounds additionally require debridement of devitalised tissue, infection control, nutritional support and, for many Stage III and IV wounds, surgical flap closure [3]. Negative-pressure wound therapy is widely used as an adjunct for deep cavity wounds.

Electrical stimulation is the adjunct with the most rehabilitation-specific evidence, and it illustrates the state of the literature well. A Cochrane review of electrical stimulation for treating pressure ulcers assessed the available trials and found the certainty of evidence limited [9]. A meta-analysis restricted to people with spinal cord injury concluded that stimulation appears to be an effective adjunct for accelerating closure, but the outcome measures across trials were so heterogeneous that the results could not be pooled into a single estimate [10]. A note of caution on the literature itself is warranted: a 2024 review of the reliability of evidence supporting electrical stimulation in this indication has since been retracted, which is a reminder that in a field with small trials and commercial interest, source quality must be assessed and not assumed.

What is not in doubt is the asymmetry of cost. Stage III and IV wounds consume the majority of the national expenditure [1], heal slowly when they heal, and recur. Treatment is the expensive failure mode of prevention, which is the strongest available argument for investing upstream.

V. Discussion and Conclusion

Read together, the four bodies of literature tell a single story. The aetiology has been re-established on mechanical rather than vascular foundations, with deformation primary and ischaemia tertiary [2]. Detection has not followed: routine practice still keys on a visible skin change that occurs after irreversible subcutaneous damage and that is systematically harder to observe in darkly pigmented skin [3], [4]. Prevention has accumulated decades of devices but mostly low-certainty evidence, in large part because trials are powered against that same late endpoint [5]–[8]. Treatment is costly, slow and incompletely evidenced [1], [9], [10].

Three implications follow for anyone designing in this space. First, **measure the precursor, not the consequence**: a device that senses tissue change days before the skin does is not an incremental improvement on visual assessment but a different measurement entirely, and it should be evaluated on lead time rather than on agreement with a lagging standard. Second, **design against all three damage contributors**. A support surface that redistributes interface pressure addresses deformation magnitude alone; it does not address shear, which drives superficial damage, nor microclimate, which lowers the deformation a tissue can tolerate. A product specified only in millimetres of mercury is specified against one third of the mechanism. Third, **treat skin-tone independence as a requirement rather than a feature**. If a detection method's performance depends on observing colour change, its accuracy varies with the patient's skin, and a measurement that is physically independent of pigmentation removes a documented disparity by construction.

The problem is therefore less a gap in treatment than a gap in observation. The most useful contribution an engineer can make is not another mattress but a reliable, inexpensive and unbiased way to see the injury while it is still preventable — which would, in turn, finally give prevention research an endpoint worth measuring.

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